Macro analysis of a BB(6) cryptid

Introduction

We study a 6-state Turing machine candidate ("cryptid") for the Busy Beaver problem. After a standard acceleration step, the machine's behaviour reduces to a macro dynamical system on three kinds of states: $P(a)$, $Q(a, b)$, and Halt. The machine halts when (and only when) the orbit reaches $Q(0, b)$ for some b. This can happen in two independent ways, depending on which Q-entry value the orbit produces:

- **Direct halt.** Q-entry values in $\{2^{i+1} - 2 : i \geq 1\} = \{2, 6, 14, 30, \ldots\}$ give rise to halving chains $Q(2^{i+1} - 2, _) \to \cdots \to Q(2, _) \to Q(0, _) = \mathsf{Halt}$ that reach $Q(0, _)$ without passing through $Q(1, _)$.
- **Via $Q(1, _)$.** Q-entry values in $\{3 \cdot 2^k - 2 : k \geq 1\} = \{4, 10, 22, 46, \ldots\}$ give rise to halving chains that pass through $Q(1, _)$, triggering rules Q1/QP1 and re-entering the P-phase with new values.

The present note focuses on the second question: the reachability of $Q(1, _)$ from the initial state $P(2)$. We develop the formal theory (verified in Lean 4 with Mathlib), giving closed forms for the P-phase map, characterising the Q-phase accumulator, and proving that $Q(1, _)$-reachability is equivalent to a Collatz-like Diophantine condition on the orbit.

1 Macro transition rules

The macro map $\mathsf{next}\colon \mathsf{State} \to \mathsf{State}$ acts by:

$$
\begin{array}{lll}
\textbf{PE:} & P(2a) & \longmapsto P(3a+4), \\
\textbf{PQ:} & P(2a{+}1) & \longmapsto Q(a{+}2,\ 1), \\
\textbf{H:} & Q(0, b) & \longmapsto \mathsf{Halt}, \\
\textbf{Q1:} & Q(1, 2b) & \longmapsto Q(b{+}2,\ 1), \\
\textbf{QP1:} & Q(1, 2b{+}1) & \longmapsto P(3b{+}8), \\
\textbf{Q2:} & Q(2a{+}2, b) & \longmapsto Q(a,\ b{+}2a{+}5), \\
\textbf{QP2:} & Q(2a{+}3, b) & \longmapsto P(b{+}5a{+}6).
\end{array}
$$

The naming convention: rules Q1, QP1 are the two cases with first argument 1; rules Q2, QP2 are the halving/exit rules with first argument ≥ 2. The orbit starts at $P(2)$.

For a state s, we write $\mathsf{next}^n(s)$ for the n-fold iterate.

Lemma 1.1 (Right-unfolding)**.** *For every state s and $n \in \mathbb{N}$,*

$$\mathsf{next}^{n+1}(s) = \mathsf{next}\big(\mathsf{next}^n(s)\big).$$

Proof. Induction on n. The base case is immediate; the step follows from the left-recursive definition of iteration: $\mathsf{next}^{n+2}(s) = \mathsf{next}^{n+1}(\mathsf{next}(s)) = \mathsf{next}(\mathsf{next}^{n+1}(s))$ by the inductive hypothesis applied to $\mathsf{next}(s)$. □

2 P-phase recurrence and closed form

When the orbit is in a P-state, rule PE applies repeatedly (each application replaces n by $3n/2+4$, stripping a factor of 2) until the argument becomes odd, at which point rule PQ fires and the orbit enters the Q-phase. The number of PE-steps equals the 2-adic valuation of the argument shifted by 8.

Definition 2.1 (P-phase map)**.** Define $p\colon \mathbb{N}_{\geq 1} \to \mathbb{N}$ by

$$p(2k+1) = k+2, \qquad p(2k) = p(3k+4) \quad (k \geq 1).$$

This recurrence terminates because $v_2(3k+4+8) < v_2(2k+8)$, where $v_2(n)$ =2-adic valuation of n.

Lemma 2.2 (Recurrence cases)**.** *For all $k \geq 1$: $p(2k) = p(3k+4)$. For all $k \geq 0$: $p(2k+1) = k+2$.*

Proof. Both follow by unfolding the definition of p and simplifying the parity check. □

The closed form expresses $p(n)$ in terms of the odd part $\mathrm{ord}_2(n+8) := (n+8)/2^{v_2(n+8)}$ and the 2-adic valuation $v_2(n+8)$.

Theorem 2.3 (Closed form for p)**.** *For $n \geq 1$,*

$$p(n) = \frac{\mathrm{ord}_2(n+8) \cdot 3^{v_2(n+8)} - 5}{2} = \frac{(n+8) \cdot 3^{v_2(n+8)} \,/\, 2^{v_2(n+8)} - 5}{2}.$$

Proof. By strong induction on $v := v_2(n+8)$.

Base ($v = 0$): Then n is odd, say $n = 2k+1$, and $p(2k+1) = k+2$. Since $v_2(2k+9) = 0$, the odd part is $2k+9$ itself, and $(2k+9) \cdot 3^0 = 2k+9$, so the formula gives $(2k+9-5)/2 = k+2$.

Step ($v \to v+1$): Then n is even, $n = 2k$, and $p(2k) = p(3k+4)$ by the recurrence. The key identities are:

$$v_2(3k+12) = v_2(k+4), \qquad \mathrm{ord}_2(3k+12) = 3 \cdot \mathrm{ord}_2(k+4),$$
$$v_2(2k+8) = v_2(k+4) + 1, \qquad \mathrm{ord}_2(2k+8) = \mathrm{ord}_2(k+4).$$

These follow from the multiplicativity of the 2-adic valuation on $3 \cdot (k+4)$ and $2 \cdot (k+4)$ (with $\gcd(3,2) = 1$). Substituting into the inductive formula for $p(3k+4)$ and simplifying yields the formula for $p(2k)$. □

Corollary 2.4 (Image of p)**.** *For every $n \geq 1$ there exist $\ell, k \in \mathbb{N}$ such that*

$$p(n) = \frac{(2\ell+1) \cdot 3^k - 5}{2}.$$

Proof. Take $\ell = \lfloor \mathrm{ord}_2(n+8)/2 \rfloor$ and $k = v_2(n+8)$. The odd part is always odd, so it has the form $2\ell+1$. □

Lemma 2.5 (Connection to macro model)**.** *For every $n \geq 1$ there exist $s \geq 0$ such that*

$$\mathsf{next}^{s+1}(P(n)) = Q(p(n),\, 1).$$

That is, iterating the macro map from $P(n)$ through the P-phase terminates at $Q(p(n), 1)$.

Proof. Induction on $v = v_2(n+8)$. If n is odd, a single PQ step gives $Q(\lfloor n/2 \rfloor + 2, 1) = Q(p(n), 1)$. If n is even, one PE step maps $P(n)$ to $P(3n/2+4)$, and the 2-adic valuation of the shifted argument decreases by 1, so the inductive hypothesis applies. □

3 Q-phase accumulator

Once the orbit enters the Q-phase at $Q(a, 1)$, rules Q2 and QP2 apply repeatedly (the first argument halves at each Q2 step, accumulating into the second argument) until the first argument becomes odd, at which point QP2 fires and returns to P. The net effect on the second argument is captured by a one-parameter accumulator.

Definition 3.1 (Q-phase accumulator)**.** Define $\mathrm{qacc}\colon \mathbb{N} \to \mathbb{N}$ by

$$\mathrm{qacc}(0) = 0, \quad \mathrm{qacc}(2k) = 2k + 3 + \mathrm{qacc}(k-1), \quad \mathrm{qacc}(2k+1) = 5k + 1.$$

Lemma 3.2 (Recurrence cases for qacc)**.** $\mathrm{qacc}(0) = 0$. *For* $k \geq 1$*:* $\mathrm{qacc}(2k) = 2k+3+\mathrm{qacc}(k-1)$. *For* $k \geq 0$*:* $\mathrm{qacc}(2k+1) = 5k+1$.

Definition 3.3 (Two-argument Q-phase function)**.** $q(a, b) := b + \mathrm{qacc}(a)$.

Remark 3.4. The accumulator qacc follows only the Q2/QP2 chain. It is well-defined for all $a \geq 0$, but at $a = 1$ it gives $\mathrm{qacc}(1) = 1$ whereas the actual Q-phase from $Q(1, b)$ would trigger rule Q1 or QP1 instead of Q2. This discrepancy is irrelevant if $Q(1, _)$ is never reached—which is the main conjecture.

Closed form for qacc

The recursion $\mathrm{qacc}(2m+2) = \mathrm{qacc}(m) + 2m + 5$ (the even case, reindexed via $k = m+1$) telescopes along the chain

$$n_0 = n, \quad n_{i+1} = \frac{n_i - 2}{2}, \quad i = 0, 1, \ldots, r-1,$$

which terminates when n_r is odd or zero. One checks by induction that $n_i = 2^{r-i}(n_r + 2) - 2$, so $n + 2 = 2^r(n_r + 2)$. Writing $v := v_2(n + 2)$ and $\mathrm{op} := \mathrm{ord}_2(n + 2)$ (the odd part), two cases arise:

- $\mathrm{op} \geq 3$**:** The chain makes $r = v$ steps, landing on odd $n_r = \mathrm{op} - 2 \geq 1$. Summing $\mathrm{qacc}(2m+2) = \mathrm{qacc}(m) + 2m + 5$ along the chain and using $\mathrm{qacc}(n_r) = (5n_r - 3)/2$:

$$\begin{aligned}\mathrm{qacc}(n) &= \sum_{i=0}^{v-1}(n_i + 3) + \mathrm{qacc}(n_v)\\ &= \mathrm{op} \cdot 2(2^v - 1) + v + \tfrac{5(\mathrm{op}-2)-3}{2}\\ &= 2n + v + \tfrac{\mathrm{op}-5}{2}.\end{aligned}$$

- $\mathrm{op} = 1$ (i.e., $n+2 = 2^v$): The chain makes $r = v-1$ steps, landing on $n_r = 0$ with $\mathrm{qacc}(0) = 0$.

$$\mathrm{qacc}(n) = 2 \cdot 2(2^{v-1} - 1) + (v - 1) = 2^{v+1} + v - 5 = 2n + v - 1.$$

These two cases unify into a single formula over $\mathbb{N}$ that avoids negative intermediate values:

Theorem 3.5 (Closed form for qacc)**.** *For* $n \geq 1$*,*

$$2\,\mathrm{qacc}(n) + 5 \;=\; 4n + 2\,v_2(n+2) + \max\bigl(\mathrm{ord}_2(n+2),\, 3\bigr).$$

Proof. By strong induction on n, following the telescoping argument above. The base case n odd: $\mathrm{qacc}(2k{+}1) = 5k + 1$, and $\mathrm{ord}_2(2k{+}3) = 2k{+}3 \geq 3$, $v_2(2k{+}3) = 0$, so the right side is $8k{+}4 + 0 + (2k{+}3) = 10k{+}7 = 2(5k{+}1) + 5$. For $n = 2m$ even with $m \geq 1$: apply $\mathrm{qacc}(2m) = 2m + 3 + \mathrm{qacc}(m{-}1)$, use $v_2(2m{+}2) = v_2(m{+}1) + 1$ and $\mathrm{ord}_2(2m{+}2) = \mathrm{ord}_2(m{+}1)$, and invoke the inductive hypothesis for $\mathrm{qacc}(m{-}1)$ (noting that $(m{-}1) + 2 = m + 1$). □

We call a Q-entry value *forbidden* if the Q2-halving chain starting from it passes through $Q(1, _)$. Since Q2 halves the first argument at each step, the chain from $Q(a, b)$ reaches $Q(1, _)$ precisely when a belongs to the backward orbit of 4 under doubling-plus-two, i.e., when $a \in \{3 \cdot 2^k - 2 : k \geq 1\} = \{4, 10, 22, 46, 94, \ldots\}$. Thus a Q-entry $Q(a, b)$ is forbidden if and only if $a = 3 \cdot 2^k - 2$ for some $k \geq 1$.

4 Backward reachability of $Q(1, _)$

The central question is whether the orbit from $P(2)$ ever reaches a state $Q(1, b)$. If it does not, then rules Q1 and QP1 are dead code, the Q-phase is always governed by Q2/QP2, and the accumulator qacc faithfully models the dynamics.

We call a P-argument value *critical* if it has the form $6 \cdot 2^j - 7$ for some $j \geq 1$, i.e., it belongs to $\{5, 17, 41, 89, 185, 377, 761, \ldots\}$. The significance of critical values is that a critical P-argument $6 \cdot 2^j - 7$ is always odd, so rule PQ fires immediately, producing $Q(3 \cdot 2^j - 2, 1)$—a forbidden Q-entry. This section proves the converse: every forbidden Q-entry in the orbit must have been preceded by a critical P-value.

We approach this by backward analysis: which states can produce $Q(1, b)$ as a successor, and how can those predecessors themselves be reached?

Lemma 4.1 (Predecessors of $Q(m, b)$ for $m \geq 2$)**.** *If* $\mathsf{next}(s) = Q(m, b)$ *with* $m \geq 2$*, then exactly one of:*

(a) $s = P(2m - 3)$ *and* $b = 1$ *(via PQ),*

(b) $s = Q(1, 2m - 4)$ *and* $b = 1$ *(via Q1),*

(c) $s = Q(2m + 2, b')$ *for some* b'*, with* $b = b' + 2m + 5$ *(via Q2).*

Proof. Exhaustive case analysis on s. If $s = \mathsf{Halt}$, the successor is Halt, contradiction. If $s = P(a)$, split on the parity of a: even gives $P(\cdots)$, which cannot equal $Q(m, b)$; odd gives $Q(a/2 + 2, 1)$, matching case (a) with $m = a/2 + 2$, i.e., $a = 2m - 3$. If $s = Q(0, b')$, the successor is Halt. If $s = Q(1, b')$, split on parity: even gives $Q(b'/2 + 2, 1)$ matching case (b); odd gives $P(\cdots)$, contradiction. If $s = Q(a' + 2, b')$, split on parity of a': even gives $Q(a'/2, b' + a' + 5)$ matching case (c) with $2m + 2 = a' + 2$; odd gives $P(\cdots)$, contradiction. □

Lemma 4.2 (Unique predecessor of $Q(1, b)$)**.** *If* $\mathsf{next}(s) = Q(1, b)$*, then* $s = Q(4, b - 7)$*, i.e., there exists* b' *with* $s = Q(4, b')$ *and* $b = b' + 7$*.*

Proof. By the same case analysis as Lemma 4.1, but specialised to $m = 1$. All cases except $Q(2 \cdot 1 + 2, b') = Q(4, b')$ lead to contradictions: the PQ case would require $P(-1)$, the Q1 case $Q(1, -2)$, and the QP cases produce P-states. □

Theorem 4.3 (Forbidden Q-entries require critical P-values)**.** *Let* $k \geq 1$*. If the orbit from* $P(2)$ *reaches a forbidden Q-entry* $Q(3 \cdot 2^k - 2, b)$ *for any* b*, then some earlier orbit state was a critical P-value: there exist* $j \geq 1$ *and* n' *such that*

$$\mathsf{next}^{n'}(P(2)) = P(6 \cdot 2^j - 7).$$

Proof. Strong induction on the orbit step n at which $Q(3 \cdot 2^k - 2, b)$ is reached.

For $n = 0$: $\mathsf{next}^0(P(2)) = P(2) \neq Q(\cdots)$, contradiction.

For $n+1$: by Lemma 1.1, let $s = \mathsf{next}^n(P(2))$ so that $\mathsf{next}(s) = Q(3 \cdot 2^k - 2, b)$. Since $3 \cdot 2^k - 2 \geq 4 \geq 2$ for $k \geq 1$, Lemma 4.1 applies, giving three cases:

(a) **PQ:** $s = P(2(3 \cdot 2^k - 2) - 3) = P(6 \cdot 2^k - 7)$. Take $j = k$.

(b) **Q1:** $s = Q(1,\, 2(3 \cdot 2^k - 2) - 4)$. By Lemma 4.2, the predecessor of this $Q(1, _)$ at step $n-1$ is some $Q(4, b'')$. Since $4 = 3 \cdot 2^1 - 2$, the inductive hypothesis applies at step $n-1$ with $k' = 1$.

(c) **Q2:** $s = Q(2(3 \cdot 2^k - 2) + 2,\, b') = Q(3 \cdot 2^{k+1} - 2,\, b')$. The inductive hypothesis applies at step n with $k' = k+1$.

In every case we obtain the desired critical P-value. □

5 The halving chain (forward direction)

The previous section showed that any forbidden Q-entry in the orbit must have been preceded by a critical P-value. Conversely, we now show that hitting a critical P-value forces the orbit into $Q(1, _)$: the PQ rule produces a forbidden Q-entry, and the halving chain then inevitably reaches $Q(1, _)$.

Lemma 5.1 (Halving chain)**.** *For $k \geq 1$ and any b, there exists b' such that*

$$\mathsf{next}^k\big(Q(3 \cdot 2^k - 2,\, b)\big) = Q(1,\, b').$$

Proof. Induction on k.

Base ($k = 1$): $Q(4, b) \xrightarrow{\text{Q2}} Q(1, b+7)$.

Step ($k+1$, with $k \geq 1$): We have $3 \cdot 2^{k+1} - 2 = 2(3 \cdot 2^k - 2) + 2$, which is even and ≥ 4, so rule Q2 fires:

$$Q\big(2(3 \cdot 2^k - 2) + 2,\, b\big) \xrightarrow{\text{Q2}} Q\big(3 \cdot 2^k - 2,\, b + 2(3 \cdot 2^k - 2) + 5\big).$$

The inductive hypothesis gives the remaining k steps to $Q(1, _)$. □

Proposition 5.2 (Critical P-values reach $Q(1, _)$)**.** *For $k \geq 1$, there exists b such that*

$$\mathsf{next}^{k+1}\big(P(6 \cdot 2^k - 7)\big) = Q(1,\, b).$$

Proof. Since $6 \cdot 2^k - 7$ is odd ($6 \cdot 2^k$ is even), rule PQ fires:

$$P(6 \cdot 2^k - 7) \xrightarrow{\text{PQ}} Q\left(\frac{6 \cdot 2^k - 8}{2} + 2,\ 1\right) = Q(3 \cdot 2^k - 2,\ 1).$$

The halving chain (Lemma 5.1) takes k more steps to $Q(1, b')$. □

6 The main conjecture and its consequences

Theorem 4.3 and Proposition 5.2 together give a complete equivalence:

> $Q(1, _)$ *is reachable from* $P(2)$ *if and only if the orbit visits a critical P-value.*

Conjecture 6.1 ($Q(1, _)$ unreachability)**.** For all $n \in \mathbb{N}$ and $b \in \mathbb{N}$,

$$\mathsf{next}^n(P(2)) \neq Q(1,\, b).$$

Equivalently, rules Q1 and QP1 are never triggered on the orbit from $P(2)$.

Corollary 6.2. *If Conjecture 6.1 holds, then:*

(i) Rule Q1 ($Q(1, 2b) \to Q(b + 2, 1)$) is never reached.

(ii) Rule QP1 ($Q(1, 2b + 1) \to P(3b + 8)$) is never reached.

Proof. Immediate from Conjecture 6.1: if $Q(1, b)$ is never reached, neither rule can fire. □

7 Arithmetic obstructions

Although the conjecture remains open, several arithmetic lemmas constrain how a critical P-value could arise in the orbit. These eliminate all but one mechanism, narrowing the problem to a single Collatz-hard condition.

Lemma 7.1 (PE cannot produce critical P-values)**.** *For all $k, j \in \mathbb{N}$ with $j \geq 1$,*

$$3k + 4 \neq 6 \cdot 2^j - 7.$$

Proof. Modular arithmetic: $3k + 4 \equiv 1 \pmod 3$, while $6 \cdot 2^j - 7 \equiv 0 - 1 \equiv 2 \pmod 3$. □

This means the PE rule $P(2a) \to P(3a + 4)$ can never produce a critical P-value. Since every critical value is odd, the only way one can appear in the orbit is as the output of a Q-phase exit (rule QP1 or QP2).

Lemma 7.2 (Closed-form Q-entries are never forbidden)**.** *For all $\ell, k, j \in \mathbb{N}$ with $k \geq 1$ and $j \geq 1$,*

$$\frac{(2\ell + 1) \cdot 3^k - 5}{2} \neq 3 \cdot 2^j - 2.$$

Proof. Assume equality. Clearing the denominator (both sides are integers since $(2\ell+1) \cdot 3^k$ is odd) gives $(2\ell + 1) \cdot 3^k = 6 \cdot 2^j + 1$. But the left side is divisible by 3 (since $k \geq 1$), while $6 \cdot 2^j + 1 \equiv 1 \pmod 3$. □

Recall from Corollary 2.4 that every Q-entry value $p(n)$ has the form $((2\ell + 1) \cdot 3^k - 5)/2$ with $k = v_2(n + 8)$. When $k \geq 1$ (i.e., n is even, so the P-phase took at least one PE step), the lemma applies and $p(n)$ is never a forbidden Q-entry. When $k = 0$ (i.e., n is odd), the PQ rule fires immediately and $p(n) = (n + 3)/2$. This equals the forbidden value $3 \cdot 2^j - 2$ precisely when $n = 6 \cdot 2^j - 7$, i.e., when n is a critical P-value. In other words: the P-phase itself (the PE iterations) never steers the orbit into a forbidden Q-entry—the only way to get one is if the P-argument was already odd and critical upon entry, which means some prior Q-phase exit must have produced that critical value.

Lemma 7.3 (QP2 output is critical iff...)**.** *The QP2 output $b+5a+6$ is a critical P-value $6 \cdot 2^j - 7$ (with $j \geq 1$) if and only if $b + 5a + 13 = 6 \cdot 2^j$.*

Proof. Immediate arithmetic (both sides integral since $6 \cdot 2^j \geq 12$). □

Remark 7.4 (Collatz-hardness). Combining the above:

- PE-outputs $3k + 4$ are never critical (Lemma 7.1).
- Q-entries $p(n)$ with $v_2(n + 8) \geq 1$ are never forbidden (Lemma 7.2).
- The only remaining source of critical P-values is the QP2 rule: $Q(2a + 3, b) \to P(b + 5a + 6)$.

Thus $Q(1, _)$ is reachable from $P(2)$ if and only if there exist a reachable state $Q(2a+3, b)$ and $j \geq 1$ with $b + 5a + 13 = 6 \cdot 2^j$, i.e., some QP2 exit produces a critical P-value. No modular invariant (checked computationally up to modulus 96) separates the orbit values from this condition. The problem has the character of a Collatz-type question: the interleaving of multiplicative (times 3) and additive structure in the P-phase, combined with the binary decomposition in the Q-phase, prevents standard density or congruence arguments from closing the proof.